

Courier and Parcel Delivery Management System

Database Design for a Real-World Application

1. Introduction

Every real-world application that stores and manages information relies on a database at its core. Before a database can be built, its structure must be planned in advance by identifying the entities (tables) involved, the attributes (columns) each entity holds, and the relationships that connect them. This assignment demonstrates that planning process for a real-world application of our choice.

2. Application Name and Purpose

Application Name: Courier and Parcel Delivery Management System

Purpose: This system is used by a courier company to manage the complete life cycle of a parcel, starting from booking at a branch, assignment to a delivery employee and vehicle, tracking of its movement between branches, and final delivery to the receiver. The system also records payments made by customers and complaints raised against a delivery, so that the company can maintain accountability and provide reliable service.

3. List of Required Tables

The following nine tables are sufficient to represent this application without unnecessary duplication of data:

- 1. Customer
- 2. Branch
- 3. Employee
- 4. Vehicle
- 5. Parcel
- 6. Delivery_Assignment
- 7. Parcel_Tracking
- 8. Payment
- 9. Complaint

4. Table Structures (Columns and Attributes)

4.1 Customer

Stores the details of every customer who books a parcel through the courier company.

Column Name	Data Type	Key	Description
CustomerID	INT	Primary Key	Unique identification number for each customer

Column Name	Data Type	Key	Description
Name	VARCHAR(100)	-	Full name of the customer
Address	VARCHAR(200)	-	Residential or office address of the customer
City	VARCHAR(50)	-	City in which the customer resides
Phone	VARCHAR(15)	-	Contact number of the customer
Email	VARCHAR(100)	-	Email address of the customer

4.2 Branch

Stores information about every branch office operated by the courier company.

Column Name	Data Type	Key	Description
BranchID	INT	Primary Key	Unique identification number for each branch
BranchName	VARCHAR(100)	-	Name of the branch office
City	VARCHAR(50)	-	City in which the branch is located
Address	VARCHAR(200)	-	Complete address of the branch
ContactNumber	VARCHAR(15)	-	Landline or mobile number of the branch

4.3 Employee

Stores the details of staff members, including delivery agents and drivers, who work at a specific branch.

Column Name	Data Type	Key	Description
EmployeeID	INT	Primary Key	Unique identification number for each employee
Name	VARCHAR(100)	-	Full name of the employee
Designation	VARCHAR(50)	-	Role of the employee, e.g. Delivery Agent, Clerk
BranchID	INT	Foreign Key	References the branch where the employee works
Phone	VARCHAR(15)	-	Contact number of the employee
DateOfJoining	DATE	-	Date on which the employee joined the company

4.4 Vehicle

Stores details of vehicles used for transporting parcels between branches and to receivers.

Column Name	Data Type	Key	Description
VehicleID	INT	Primary Key	Unique identification number for each vehicle
VehicleNumber	VARCHAR(20)	-	Registration number of the vehicle
VehicleType	VARCHAR(30)	-	Type of vehicle, e.g. Van, Two-Wheeler, Truck
BranchID	INT	Foreign Key	References the branch the vehicle is stationed at

4.5 Parcel

The central table of the system. Stores every parcel booked by a customer, along with sender, receiver and routing details.

Column Name	Data Type	Key	Description
ParcelID	INT	Primary Key	Unique identification number for each parcel
CustomerID	INT	Foreign Key	References the customer who booked (sent) the parcel
ReceiverName	VARCHAR(100)	-	Name of the person receiving the parcel
ReceiverAddress	VARCHAR(200)	-	Delivery address of the receiver
ReceiverPhone	VARCHAR(15)	-	Contact number of the receiver
Weight	DECIMAL(6,2)	-	Weight of the parcel in kilograms
ParcelType	VARCHAR(30)	-	Category of parcel, e.g. Document, Fragile, Package
BookingBranchID	INT	Foreign Key	References the branch where the parcel was booked
DestinationBranchID	INT	Foreign Key	References the branch nearest to the receiver
BookingDate	DATE	-	Date on which the parcel was booked
DeliveryStatus	VARCHAR(20)	-	Current overall status, e.g. Pending, Delivered

4.6 Delivery_Assignment

Links a parcel to the employee and vehicle responsible for delivering it. A new record is created if a parcel needs to be reassigned for a second delivery attempt.

Column Name	Data Type	Key	Description
AssignmentID	INT	Primary Key	Unique identification number for each assignment
ParcelID	INT	Foreign Key	References the parcel being delivered
EmployeeID	INT	Foreign Key	References the employee assigned for delivery
VehicleID	INT	Foreign Key	References the vehicle used for delivery
AssignedDate	DATE	-	Date on which the assignment was made
ExpectedDeliveryDate	DATE	-	Date by which the parcel is expected to be delivered
ActualDeliveryDate	DATE	-	Date on which the parcel was actually delivered

4.7 Parcel_Tracking

Stores the movement history of a parcel so that both the company and the customer can track its current location and status at any time.

Column Name	Data Type	Key	Description
TrackingID	INT	Primary Key	Unique identification number for each tracking entry
ParcelID	INT	Foreign Key	References the parcel being tracked
CurrentLocation	VARCHAR(100)	-	Location of the parcel at the time of the update
Status	VARCHAR(30)	-	Status such as Booked, In Transit, Out for Delivery, Delivered
StatusDateTime	DATETIME	-	Date and time at which this status was recorded

4.8 Payment

Stores payment details for each parcel booking, since every parcel is charged based on weight and type.

Column Name	Data Type	Key	Description
PaymentID	INT	Primary Key	Unique identification number for each payment
ParcelID	INT	Foreign Key	References the parcel this payment was made for
AmountPaid	DECIMAL(8,2)	-	Amount paid by the customer
PaymentMode	VARCHAR(20)	-	Mode of payment, e.g. Cash, Card, UPI
PaymentDate	DATE	-	Date on which the payment was made
PaymentStatus	VARCHAR(20)	-	Status of payment, e.g. Paid, Pending

4.9 Complaint

Stores complaints raised by customers regarding a specific parcel, such as delay or damage, so the company can track and resolve them.

Column Name	Data Type	Key	Description
ComplaintID	INT	Primary Key	Unique identification number for each complaint
ParcelID	INT	Foreign Key	References the parcel the complaint is about
CustomerID	INT	Foreign Key	References the customer who raised the complaint
ComplaintDate	DATE	-	Date on which the complaint was registered
Description	VARCHAR(250)	-	Details of the complaint provided by the customer
ComplaintStatus	VARCHAR(20)	-	Status of the complaint, e.g. Open, Resolved

5. Relationships Between Tables

The tables are connected to one another through primary key and foreign key relationships, as summarised below:

Tables Involved	Relationship Type	Explanation
Branch - Employee	One-to-Many	One branch employs many employees, but each employee works at only one branch.
Branch - Vehicle	One-to-Many	One branch has many vehicles stationed at it, but each vehicle belongs to one branch.
Customer - Parcel	One-to-Many	One customer can book many parcels, but each parcel is booked by one customer.
Branch - Parcel	One-to-Many	Each parcel has one booking branch and one destination branch; a branch can appear on many parcels in either role.
Parcel - Delivery_Assignment	One-to-Many	A parcel usually has one assignment, but may have more than one if a redelivery attempt is required.
Employee - Delivery_Assignment	One-to-Many	One employee can be assigned to deliver many parcels over time.
Vehicle - Delivery_Assignment	One-to-Many	One vehicle can be used for many delivery assignments over time.
Parcel - Parcel_Tracking	One-to-Many	One parcel passes through many tracking updates as it moves from

Tables Involved	Relationship Type	Explanation
		booking to delivery.
Parcel - Payment	One-to-One	Each parcel has exactly one corresponding payment record.
Customer - Complaint	One-to-Many	One customer may raise complaints for more than one parcel.
Parcel - Complaint	One-to-Many	A parcel may have more than one complaint raised against it, e.g. delay followed by a damage complaint.

6. Purpose of Each Table (Summary)

Table Name	Purpose
Customer	Maintains the record of individuals who send parcels through the company.
Branch	Maintains the record of every physical branch office of the company.
Employee	Maintains staff records, including which branch each employee belongs to.
Vehicle	Maintains the record of vehicles available for parcel delivery.
Parcel	Central table that holds sender, receiver and routing details of every parcel booked.
Delivery_Assignment	Records which employee and vehicle were assigned to deliver a specific parcel.
Parcel_Tracking	Maintains a running history of a parcel's location and status for tracking purposes.
Payment	Maintains billing information for each parcel that is booked.
Complaint	Maintains customer complaints so that service issues can be resolved and monitored.

7. Conclusion

The Courier and Parcel Delivery Management System can be effectively represented using nine interconnected tables: Customer, Branch, Employee, Vehicle, Parcel, Delivery_Assignment, Parcel_Tracking, Payment and Complaint. Each table stores one distinct type of information required by the application, and the relationships between them, established through primary and foreign keys, ensure that data such as bookings, deliveries, tracking history, payments and complaints remain organised and easy to retrieve. This design avoids redundant tables while still covering every core function the courier company needs to operate.